

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our newly developed predictive model demonstrates strong performance, accurately forecasting outcomes for over 83% of cases. This high level of reliability provides a solid foundation for understanding the core influences behind our objectives, enabling more informed strategic and operational decision-making. Analysis of the model's insights reveals three critical business drivers. Gender, specifically the male demographic, consistently emerges as the most powerful determinant of the predicted outcome. The amount paid (fare) is another highly significant factor, alongside an individual's age, which also plays a crucial role in influencing results. These elements are the primary forces shaping our observed outcomes. Given these key findings, it is imperative to closely monitor these influential demographic and transactional variables. Any strategies or operational planning related to the outcome should pay particular attention to gender, fare, and age, as shifts in these areas are likely to have the most substantial impact on our business performance. Regularly tracking these drivers will be essential for proactive management and effective adaptation.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

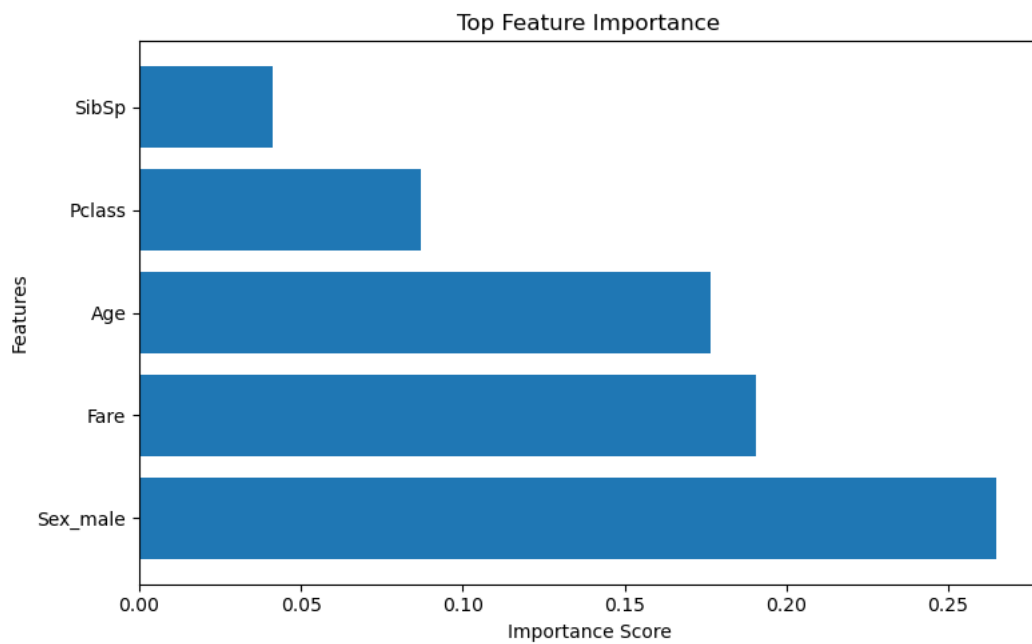
- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.
- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

